

Gold Evaluation & Re-Validation

Step 7 — Do Steps 2–5’s marker-based findings hold up on real gold?

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1. Objective

Steps 2–5 evaluated Task 1 (clustering) and Task 2 (linking) against *marker-derived* canonical topics (151 topics, only 30 recurring) — a proxy for gold, not gold itself. Step 6 produced real full-protocol gold segmentation (523 segments, 211/213 protocols). This step converts that into gold data for both tasks and re-runs every prior evaluation against it, to check whether the earlier conclusions (best representation, best hybrid weight, expected streaming performance) actually hold at realistic granularity.

2. Method

`build_gold.py` converts `segmentation.json` into `gold_segments.json` (523 segments) and `gold_chains.json` (topic string → chronological occurrence list; 22 keys have ≥ 2 occurrences = recurring). `embed_gold.py` computes e5 and AlephBERT embeddings of every gold span on GPU. `gold_reeval.py` re-runs Task 1 clustering (ARI/AUC) and Task 2 pairwise + streaming linking (precision/recall/F1) against this gold, exactly mirroring Steps 3–5’s methodology. `gold_abtt_and_hybrid.py` re-sweeps ABTT k and the hybrid α weight to check whether Step 3–4’s tuned values still hold.

3. Results

3.1 Task 1 — Clustering (58 recurring gold segments, 22 gold topics)

Representation	ARI (agglom.)	ARI (kmeans)	AUC
TF-IDF	0.382	0.407	0.874
e5_k1	0.657	0.515	0.954
AlephBERT $k=2$	0.265	0.274	0.854
Hybrid $\alpha=0.6$	0.265 (spectral 0.411)	—	0.884

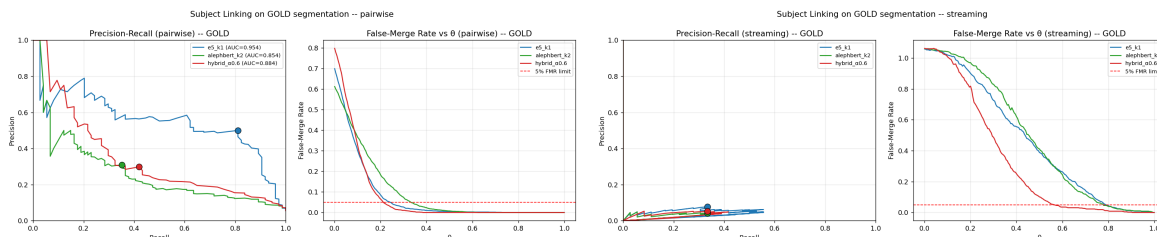
3.2 Task 2 — Linking

Pairwise (oracle): e5_k1 best F1= 0.62 @ $\theta=0.26$, AUC= 0.954 — all higher than the Step 3/5 marker-based estimates.

Streaming (realistic) — the headline result:

Representation	Best F1	Precision at that point	Recall
e5_k1	0.125	7.7%	33%
AlephBERT $k=2$	0.075	4.2%	33%
Hybrid $\alpha=0.6$	0.090	5.2%	33%

Streaming F1 collapsed from Step 5’s marker-based estimate of 0.41 to 0.125 on real gold. At the best threshold, roughly 92% of “LINK” predictions are false merges.



4. Key Finding

Not a representation problem. e5_k1 is still the best representation tried, by an even wider margin than on marker-based data. **It is a task-formulation problem:** gold segmentation is far more fine-grained (523 segments, only 22 ever recur) than the old canonical topics (30/151 recurring), so true “same subject” pairs are rare, and any similarity threshold loose enough to catch them is loose enough to flood in unrelated same-sounding topics. No fixed cosine cutoff can be both loose enough for recall and tight enough for precision at this granularity.

Follow-up checks ruled out easy fixes: Step 4’s hybrid finding does not replicate on gold (hybrid is worse than plain e5_k1 on every clustering metric), and re-sweeping ABTT k confirms e5 $k=1$ is still $\approx$ optimal.

5. What Needs Improvement

1. **The core open problem:** no embedding-threshold method can fix the precision ceiling demonstrated here. This directly motivated Step 8 (retrieve-then-verify with an LLM), which is itself still below this baseline as of writing — see that report.
2. **No representation tuning left to try.** e5_k1 unchanged is confirmed as the representation to carry forward; further ABTT/hybrid sweeps are not expected to help.
3. **Gold chain undercount.** As noted in Step 6, no label canonicalization was applied, so the “22 recurring topics” figure is likely an undercount of true recurrence in the corpus — some real repeats are hidden as singleton chains under slightly different generated labels.

Artifacts (this step):

outputs/gold_segments.json, outputs/gold_chains.json,
 outputs/gold_reeval_results.json, outputs/gold_abtt_sweep.json,
 outputs/gold_hybrid_clustering.json, outputs/embeddings/
Code: code/build_gold.py, code/embed_gold.py, code/gold_reeval.py,
 code/gold_abtt_and_hybrid.py.